Can Reinforcement Learning effectively detect money laundering, and how does it compare to leading supervised learning methods on synthetic transaction data?

Kaan Gogcay
Hogeschool van Amsterdam

March 16, 2026

Abstract

Money laundering is a significant global problem, with an estimated 16 billion euros laundered annually in the Netherlands alone. While machine learning has been widely adopted for Anti-Money Laundering (AML) transaction monitoring, the application of Reinforcement Learning (RL) to this domain remains largely unexplored. Existing RL approaches either rely on graph-based methods not applicable to traditional banking data, or target credit card fraud specifically rather than money laundering. This research investigates whether a Deep Q-Network (DQN) can effectively detect money laundering on tabular banking transaction data, and how it compares to XGBoost, the current industry and academic standard. The SAML-D dataset, a synthetic AML dataset developed in consultation with AML specialists, is used for training and evaluation. Both models are evaluated using the Area Under the Precision-Recall Curve (AUPRC), consistent with industry practice at ING and ABN AMRO, and SHAP values are applied to interpret model decisions in accordance with the explainability requirement imposed by the Dutch Anti-Money Laundering Act (Wwft). The research follows an experimental design with a chronological 70/15/15 train/validation/test split to prevent data leakage. This work contributes to the literature by being the first to apply pure Reinforcement Learning to AML detection on tabular banking transaction data, with legally required explainability incorporated into the proposed model.

1 Introduction

1.1 Context Analysis

1.1.1 Money Laundering

Money laundering is the process of concealing the illegal origin of funds to make them appear legitimate, allowing criminals to use the proceeds in the legal economy [10, 9]. These funds often come from crimes such as drug trafficking, fraud, or illegal pornography, tax evasion, robbery, blackmail, bribery, piracy, people trafficking and many more malicious activities [10, 5, 14]. Once obtained, the money cannot be spent freely, so criminals must disguise its illegal origin. This is achieved through the money-laundering process, which typically occurs in three stages: placement, layering and integration [5].

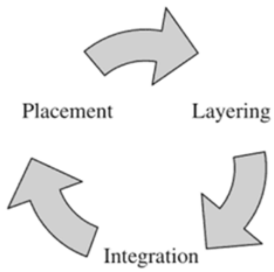

Figure 1: The money-laundering process

Placement Placement is the first step in the laundering process. In this step, criminals attempt to introduce their funds into the financial system. This is often associated with depositing cash into bank accounts, but placement is not limited to plain deposits. Some examples of introducing criminal money to the financial system cover: purchasing artwork, antiques and lottery tickets.

Layering Once criminally earned funds have been introduced to the financial system, criminals try to hide the origin of this money. This can be done in various ways, such as investing in legitimate assets, buying and selling collectibles, or making multiple smaller transactions across different accounts. The more these funds are moved back and forth, the harder it becomes to trace their original source.

Integration Once the origin of illegal funds is difficult to trace, criminals can freely integrate their laundered money into the economy. This can be done by investing in businesses, purchasing property, buying jewellery, or other assets that appear legitimate.

1.1.2 Money Laundering Detection

Money laundering detection or AML (anti-money laundering) refers to the process of detecting money laundering. AML technology has changed significantly over the past years. Evolving from dedicated AML teams to semi-automated machine learning workflows [20].

1.1.3 Limitations of Traditional AML Methods

Rule-based systems became one of the first forms of automated detection and are still used in current day [3, 14]. These systems typically monitor transactions and generated alerts for activity exceeding thresholds, such as amounts over $10,000, amounts just below $10,000, or multiple smaller transactions within 24 hours that collectively surpassed $10,000 [20]. Over time, banks have expanded the amount of rules in their systems, in attempt to catch as many suspicious transactions as possible. ING, a major bank in the Netherlands, reported using close to 500 rules as of March 2026 [3]. While rule-based systems improved efficiency, they had limitations in handling complex laundering patterns and large-scale data, which led to the adoption of machine learning [14].

1.2 Literature Overview

This section provides an overview of the current state-of-the-art regarding Machine Learning applications within Anti-Money Laundering.

1.2.1 State of the Art

In 2019, Kaggle hosted the *IEEE-CIS Fraud Detection Competition* in collaboration with Vesta Corporation, an event that remains a foundational milestone for machine learning in the financial sector [11]. By providing a massive, real-world dataset directly from Vesta's e-commerce transactions, the competition set a definitive benchmark for the state-of-the-art in transaction monitoring and fraud detection algorithms.

Vesta, a leader in e-commerce payment solutions, shared this data to challenge researchers to improve the effectiveness of fraudulent transaction alerts. The competition consisted of two main goals: to increase the detection of criminal activity while minimizing the "false positives" that lead to legitimate customers having their cards declined at the checkout.

The competition allowed researchers to join individually or in teams, with the flexibility to submit an unlimited number of solutions to refine their results. The scale of the competition is reflected in the final statistics: 7,389 individual researchers, 6,351 unique teams, 125,219 submissions and a $20,000 prize pool

The competition results provide a clear map of what defines as the "state-of-the-art" for tabular fraud data. An analysis of the top-performing teams (which shared their solution) reveals a heavy reliance on supervised learning, specifically Gradient Boosting Decision Trees (GBDT).

- **1st Place (Score: 0.9677)**: The winning solution utilized a stacked ensemble of CatBoost, XGBoost, and LightGBM. The team also developed a Neural Network, but it was dropped from the final submission because it underperformed compared to the Gradient Boosting models [7].
- **2nd Place (Score: 0.9672)**: This team achieved their ranking using a blend of LightGBM, CatBoost, and XGBoost. Their performance was further boosted by a temporal post-processing step that overrode predictions based on a user's past fraud history [4].
- **5th Place (Score: 0.9424)**: The 5th place solution relied on a simple average of four independent LightGBM models. They also created a specialized model focused on "single transaction" users, the output of which was used as a feature for their main LightGBM ensemble [18].

1.2.2 More Recent Developements

A recent survey (2017–2023) analyzed supervised AML models trained on real-world financial data, summarizing results across studies into a comparative overview [22].

As shown in [Table 1, [22]], the most frequent used supervised methods in this time period are Logistic Regression (6 studies), Random Forest (5), XGBoost (4), Decision Tree (3) and Support Vector Machines (3). The table also indicates what models have been rarely explored recently (only once) on real banking data, namely: Artificial Neural Networks, Gru, LSTM, Transformers, Decision Regression Tree, Naive Bayes, K-Nearest Neighbors, Feedforward Network, AdaBoost, Graph Deep Learning and Graph Convolutional Network. It is worth noting that the last three of these rarer methods were only applied to cryptocurrency transaction datasets rather than traditional banking data, leaving their potential on traditional banking data unexplored.

1.2.3 Developments within Reinforcement Learning

To date, Reinforcement Learning has barely been explored within AML.

One paper proposes FraudGNN-RL [6], which combines GNNs with RL for adaptive fraud detection. Transactions are modeled as a dynamic graph and a Deep Q-Network dynamically adjusts the detection threshold to adapt to evolving fraud patterns. Experiments on a real-world dataset achieved a 97.3% F1-score and 31% fewer false positives compared to baselines including XGBoost, GCN, and several other methods.

Another paper [17] also explores RL utilizing a Deep Q-Network, but without the use of graph-based methods. The model was trained and tested on a publicly available Kaggle credit card fraud dataset, achieving 97.1% validation accuracy. It is worth noting that this paper focuses on credit card fraud rather than money laundering specifically, though both problems are fundamentally based on transaction monitoring.

1.3 Stakeholder Insights

To ground this research in real-world practice, presentations were attended from data scientists at two major Dutch banks: ING [3] and ABN AMRO [14].

1.3.1 ING

ING uses a workflow including a rule-based system and a human-in-the-loop to generate labels for the supervised models [3]. ING currently operates approximately 500 static rules to flag potentially suspicious transactions. When a rule fires, a human investigator reviews the transaction and determines whether it is genuinely suspicious. This judgment then becomes the label. This labeling process is therefore inherently dependent on both the quality of the existing rules and human judgment.

ING exclusively uses supervised classification models for transaction monitoring. Tree-based models, particularly XGBoost, being the preferred approach due to their strong performance and explainability. The data scientists would like to explore more complex models such as neural networks and graph-based approaches but face practical barriers, including engineering constraints given the scale of the data (approximately 8 billion transactions) and the difficulty of obtaining regulatory sign-off for less interpretable models. Explainability is a must and is therefore not negotiatable. Also worthy to mention is that the compliance team does not permit training on synthetic data.

Given that potential financial crime represents only 1–2% of all transactions, the data is highly imbalanced. Rather than addressing this through resampling, ING uses the Area Under the Precision-Recall Curve (AUPRC) as their primary evaluation metric, a measure well suited to imbalanced data, targeting a minimum recall of 0.95. Accuracy is explicitly avoided as a standalone metric as it would be misleading under such class imbalance.

1.3.2 ABN AMRO

ABN AMRO takes a broader approach than ING, combining both supervised and unsupervised models within their transaction monitoring pipeline [14]. Each transaction passes through two parallel flows: one using supervised models, including XGBoost, decision trees, and random forests, to detect known money laundering patterns, and another using unsupervised models, including auto-encoders, clustering, and isolation forest, to detect unknown or anomalous behaviour. Both flows feed into manual review by investigators before anything is reported to the FIU.

Similar to ING, labels at ABN AMRO originate from rule-based systems, a process they acknowledge is error-prone. They expressed a desire to move towards ML-based labelling in the future.

A notable practice at ABN AMRO is their operational use of SHAP values. Every alert generated by the model is accompanied by a ranked list of the top three features that contributed to the alert, along with a non-technical explanation for investigators. This confirms that SHAP is not only theoretically suitable for AML explainability but is already used in production at a major Dutch bank.

Similar to ING, ABN AMRO uses AUPRC as their primary evaluation metric. The detection threshold is not fixed by the data scientists alone. It is determined in consultation with business stakeholders, balancing the desire to catch as many cases as possible against the number of alerts investigators can realistically review.

1.3.3 Comparison

Both banks confirm that tree-based models, particularly XGBoost, represent the current state of the art in production AML systems, being consistent with the literature. Both face the same fundamental labelling challenge, where ground truth is derived from rule-based systems rather than verified financial crime cases. The key difference is that ABN AMRO combines supervised and unsupervised approaches, while ING relies exclusively on supervised models. Both banks use AUPRC as their primary metric, which sets the evaluation strategy of this research.

1.4 Identified Gap

Despite the promise of Reinforcement Learning in adaptive fraud detection, its application to AML on tabular transaction data remains largely unexplored. The literature reveals several concrete gaps.

First, graph-based methods such as Graph Convolutional Networks and Graph Deep Learning, which are well suited to the relational nature of banking transactions, have only been applied to cryptocurrency datasets such as the Elliptic dataset [1, 21]. Their potential on traditional banking transaction data remains unexplored.

Second, while Reinforcement Learning has shown promise in transaction monitoring, existing RL approaches have not been applied to AML specifically. FraudGNN-RL combines GNN with RL but was evaluated on a general fraud dataset rather than a dedicated AML dataset [6]. The only pure RL approach found in the literature targets credit card fraud, which, while related, is a fundamentally different problem from money laundering [17].

Taken together, these gaps point to a clear research opportunity: applying Reinforcement Learning to AML on tabular transaction data, without relying on graph structure that may not be available in practice.

1.5 Research Question

Based on the identified gap, this research investigates the following main question:

> *Can Reinforcement Learning effectively detect money laundering, and how does it compare to leading supervised learning methods on synthetic transaction data?*

To answer this, the following subquestions are addressed:

1. How can Reinforcement Learning be applied to AML transaction monitoring?
2. What data is suitable for evaluating Reinforcement Learning in AML?
3. How does Reinforcement Learning perform compared to leading supervised learning methods on this data?
4. What are the benefits and limitations of Reinforcement Learning for AML based on these results?

1.6 Scope

This research focuses on applying Reinforcement Learning to AML transaction monitoring using tabular data. Although the literature highlights graph-based methods such as GCN and Graph Deep Learning as promising directions, these are explicitly out of scope.

Since the context of this research involves banking institutions as stakeholders, it may be assumed that bank data is used. This is not the case. Due to privacy and legal constraints, this research relies exclusively on publicly available AML datasets. Cryptocurrency transaction datasets are out of scope.

Furthermore, while the RL papers discussed in the literature review originate from the broader fraud detection domain, this research is scoped specifically to money laundering detection. Credit card fraud and other forms of financial fraud are related but distinct problems and are not the subject of this research.

Finally, given that the Wwft requires financial institutions to be able to explain AML decisions, explainability of the RL model is addressed in this research [15]. However, a full legal compliance audit is out of scope. The explainability analysis serves to assess whether the model's decisions can be interpreted in a meaningful way, not to provide a legal judgment on its deployability.

2 Background

2.1 Relevant Literature on Techniques

2.1.1 Reinforcement Learning and Deep Q-Networks

Reinforcement Learning (RL) is a machine learning paradigm in which an agent learns to make decisions by interacting with an environment. Rather than learning from a labeled dataset, the agent receives a reward or penalty based on its actions and uses this feedback to improve its policy over time. This makes RL particularly suitable for sequential decision-making problems where the optimal action depends on context and may change over time. A widely used RL algorithm is the Deep Q-Network (DQN), which combines Q-learning with a neural network to approximate the value of taking a given action in a given state. DQN has been successfully applied to transaction monitoring in both [6] and [17], making it the most prominent RL approach in this domain and a natural starting point for this research.

2.1.2 Explainability with SHAP

SHAP (SHapley Additive exPlanations) is an explainability method that can be applied to any machine learning model, assigning each input feature a contribution score for a given prediction. This makes it particularly suitable for interpreting black-box models such as the Deep Q-Network used in this research. In the context of AML, SHAP has already been applied in [19] (also discussed in the survey by [22]) and by ABN AMRO [14] to identify which features contributed most to a model flagging a customer as high-risk. Given the explainability requirements imposed by the Wwft, SHAP will be used in this research to interpret the decisions made by the RL model.

2.2 Related Work Comparison

As established in the literature overview, only two papers were found that apply Reinforcement Learning to transaction monitoring. Table 1 provides a structured comparison between these papers and the research proposed in this thesis.

Table 1: Comparison of related work

	Cui et al. [6]	**Qayoom et al. [17]**	**This research**
Method	GNN + RL	RL only	RL only
Domain	General fraud	Credit card fraud	AML
Data	PaySim, Credit Card 2023, IEEE-CIS	ULB Credit Card Fraud (Kaggle)	TBD
Metrics	AUC-ROC, AUC-PR, F1, Recall@k%	Accuracy and loss only	AUPRC
Explainability	No	No	Yes (SHAP)

The comparison highlights that while both related papers apply Reinforcement Learning to transaction monitoring, neither targets money laundering specifically. FraudGNN-RL relies on graph structure that may not be available in traditional banking environments, while Qayoom et al. focus on credit card fraud, which is a fundamentally different problem from money laundering.

A notable methodological concern with Qayoom et al. is their exclusive use of accuracy and loss as evaluation metrics. Given that their dataset is highly imbalanced (with only 0.172% fraudulent transactions) accuracy alone is a misleading indicator of model performance. A model that classifies every transaction as legitimate would still achieve over 99% accuracy. Metrics such as AUC-ROC, AUC-PR, precision, recall, and F1-score are better suited for imbalanced fraud detection tasks, as used by FraudGNN-RL. This research will therefore adopt AUPRC as its evaluation metric, consistent with industry practice at both ING and ABN AMRO [3, 14].

2.3 AI Suitability Justification

The suitability of AI for AML transaction monitoring is supported by both the literature and insights gathered from presentations by data scientists at ING [3] and ABN AMRO [14].

The fundamental limitation of rule-based systems is that they assume the programmer has a complete picture of all criminal patterns. As ABN AMRO noted, laundering patterns are too diverse and dynamic for rules to cover every scenario; at some point, the number of rules simply becomes unmanageable [14]. ING's reported use of approximately 500 static rules proves this problem in practice [3]. Machine learning addresses this by learning complex patterns directly from data, allowing the system to detect suspicious behaviour without relying on predefined thresholds.

A further argument for AI is the nature of money laundering itself. As ABN AMRO observed, individual transactions may not appear suspicious by itself, it is the patterns across transactions that reveal criminal activity [14]. Machine learning models are well suited to capture these patterns, particularly at the layering stage of the laundering process, which is both the most difficult stage for rules to catch and the most suitable to model-based detection [14].

Both banks also confirmed that tree-based models, particularly XGBoost, currently outperform more complex alternatives in practice, not because simpler models are theoretically superior, but because they offer the best trade-off between performance and explainability given regulatory constraints [3, 14]. This confirms that AI is not only suitable for AML but is already the industry standard, while also highlighting that any proposed model must remain interpretable to satisfy regulators.

2.4 Ethical and Societal Considerations

The application of machine learning to AML transaction monitoring raises several ethical and societal considerations that are relevant to this research.

2.4.1 Societal Impact

Money laundering has a significant societal cost. ABN AMRO estimated that approximately 16 billion euros is laundered in the Netherlands annually, placing the country eighth globally in terms of laundered funds [14]. Effective AML detection therefore has a direct positive impact on society by disrupting criminal activity and protecting the integrity of the financial system.

2.4.2 False Positives and Customer Impact

A key ethical concern in AML systems is the consequences of false positives. When a legitimate customer is incorrectly flagged as suspicious, they may face restricted access to banking services, delayed transactions, or reputational harm. Both ING and ABN AMRO explicitly aim to reduce false positives in their systems [3, 14]. This research adopts the same objective by optimizing on AUPRC, which balances precision and recall.

2.4.3 Bias and Fairness

Machine learning models trained on historical transaction data risk inheriting biases present in that data. For example, if certain demographic groups or transaction types were historically flagged more often, a model may learn to replicate these patterns. ING acknowledged the existence of dedicated teams for bias and fairness assessment [3], reflecting the seriousness of this concern.

2.4.4 Labeling Bias

A subtler but important issue is that the labels used to train supervised AML models are themselves derived from rule-based systems and human investigators. As both banks acknowledged, this process is error-prone [3, 14]. A model trained on these labels may therefore learn to replicate the limitations and biases of the existing system rather than detecting true financial crime. This is a fundamental challenge in AML that this research inherits and acknowledges as a limitation.

2.4.5 Explainability as an Ethical Requirement

Explainability is not only a legal requirement under the Wwft [15] but also an ethical one. When a model flags a customer as suspicious, investigators and potentially customers themselves deserve to understand why. ABN AMRO's operational use of SHAP values to provide investigators with a ranked explanation of each alert reflects this principle in practice [14]. This research adopts the same approach by incorporating SHAP-based explainability into the proposed model.

2.4.6 Data Privacy

The use of personal financial data raises privacy concerns. In the Netherlands, the GDPR (known locally as the AVG) restricts what personal data can be used for model training [8]. ING noted that sensitive attributes such as name, address, gender, nationality, and country of origin are not accessible for model training [3]. This research addresses privacy concerns by relying exclusively on publicly available, anonymized datasets rather than real customer data.

3 Methodology: Reserach Design

3.1 Planned Research Design

This research follows an experimental design in which a Reinforcement Learning model and a supervised learning baseline are implemented, trained, and evaluated on the same AML dataset. The research proceeds in four phases. First, a suitable publicly available AML dataset is selected and preprocessed. Second, a baseline supervised model is implemented based on the current state of the art in both the literature and industry practice. Third, a Deep Q-Network is implemented and applied to the same AML detection task. Finally, both models are evaluated using the same metrics and validation strategy, and their results are compared to answer the main research question.

3.2 Type of Study

This research is an empirical, experimental study involving the construction and evaluation of machine learning models on financial data. This approach is necessary because, as established in the literature overview, RL has not previously been applied to AML on tabular banking data.

3.3 Baseline Definition

The baseline model is XGBoost, an extreme gradient boosting algorithm. This choice is justified on three grounds. First, the academic literature consistently identifies XGBoost as one of the most frequently used methods in AML, appearing in four of the studies surveyed by [22]. Second, the top-performing teams in the IEEE-CIS Fraud Detection Competition [11] relied heavily on XGBoost as part of their winning solutions [7, 4, 18], establishing it as the state of the art for tabular fraud detection. Third, both ING and ABN AMRO confirmed in their presentations that XGBoost is their primary model in production [3, 14]. Using XGBoost as the baseline therefore ensures that the RL model is benchmarked against a method that is both academically established and practically relevant.

3.4 Comparison Logic

Both models will be trained and evaluated on the same dataset, using the same train-test split to ensure a fair comparison. The primary evaluation metric is AUPRC, consistent with industry practice at ING and ABN AMRO [3, 14]. AUPRC is particularly suitable for imbalanced datasets, where money laundering cases represent only a small fraction of all transactions. A model is considered to outperform the baseline if it achieves a higher AUPRC score under the same experimental conditions.

4 Requirements

4.1 Legal & Ethical Requirements

This research operates within the Dutch legal framework for financial crime detection and data privacy. This highlights two particular laws: *Wet ter voorkoming van witwassen en financieren van terrorisme (Wwft)*, the Dutch implementation of the EU Anti-Money Laundering Directive [15] and *General Data Protection Regulation (GDPR)*, known locally as Algemene Verordening Gegevensbescherming (AVG) [8].

Table 2: Legal & Ethical Requirements

ID	Requirement	Source
LERQ1	When a transaction is flagged as suspicious, the model must be able to explain why.	Wwft, Art. 16.2e [15]
LERQ2	The use of the following personal attributes is explicitly prohibited in model training: race, ethnic background, political opinion, religious or philosophical beliefs, trade union membership, genetic data, biometric data, health data, and data concerning a natural person's sexual behaviour or sexual orientation.	GDPR, Art. 9.1 [8]

4.2 Functional Requirements

The following functional requirements define what the model must be capable of doing:

Table 3: Functional Requirements

ID	Requirement	Source
FR1	The model must classify transactions as either suspicious or legitimate.	Research scope
FR2	The model must output a continuous suspiciousness score per transaction.	Required for AUPRC, Industry practice [3]
FR3	The model must provide a feature-level explanation for each flagged transaction using SHAP values.	Wwft, Art. 16.2e [15];
FR4	The model must be evaluated using AUPRC as the primary metric.	Industry practice [3, 14]
FR5	Both the RL model and the XGBoost baseline must be trained and evaluated on the same dataset under the same conditions to ensure a fair comparison.	Research scope

4.3 Technical Requirements

The following technical requirements define how and in what circumstances the model must be built:

Table 4: Technical Requirements

ID	Requirement	Source
TR1	The dataset must be tabular	Research scope
TR2	The dataset must be anonymized, containing no personal identifiers.	GDPR, Art. 9.1 [8]
TR3	The dataset must not contain cryptocurrency transactions.	Research Scope
TR4	A Deep Q-Network architecture must be used for the RL-model	Related Work [6, 17]
TR5	The baseline model must be implemented using XGBoost.	Literature [22] and Industry practice [3, 14]

5 Methodology: Data

5.1 Data Sources

Real-world AML datasets are not publicly available [12]. Therefore, multiple synthetic datasets have been considered for this research, as shown in Table 5. Note that *IBM AML* appears six times in the table since it provides six variations of its dataset [2]. It is also worth noting that both *SAML-D* and *money laundering data* provide a data generator, meaning that if the data volume or class imbalance of the existing dataset is not suitable, additional data can be generated as needed [16, 13].

SAML-D stands out for having a perfect kaggle score, meaning it is easy to use and well-documented. Also does it have the highest amount of features compared to the other datasets and a high data volume of approximately 9.5 million transactions. For these reasons, SAML-D is selected as the dataset for this research.

Table 5: AML Dataset Comparison

Dataset	**Rows**	**Features**	**Type**	**Labels**	**Imbalance (% fraud)**	**Kaggle Score**
IBM AML (HI-Large)	180M	11	Synthetic	Yes	0.125%	9.41
IBM AML (HI-Medium)	32M	11	Synthetic	Yes	0.110%	9.41
IBM AML (HI-Small)	5M	11	Synthetic	Yes	0.100%	9.41
IBM AML (LI-Large)	176M	11	Synthetic	Yes	0.057%	9.41
IBM AML (LI-Medium)	31M	11	Synthetic	Yes	0.050%	9.41
IBM AML (LI-Small)	7M	11	Synthetic	Yes	0.051%	9.41
SAML-D	9.5M	12	Synthetic	Yes	0.104%	10.00
SynthAML	20,000	2	Synthetic	Yes	17.175%	N/A
money laundering data	2,340	7	Synthetic	Yes	59.786%	7.65

5.2 Data Characteristics

The SAML-D dataset [16] is a tabular dataset consisting of 9,504,852 transactions across 12 features, stored in CSV format. The target column `Is_laundering` is a binary indicator distinguishing normal from suspicious transactions.

5.2.1 Features

The dataset contains the following features, grouped into five categories:

Table 6: SAML-D Feature Overview

Category	**Feature**	**Description**
Temporal	`Time`	Time of transaction (hh:mm:ss)
	`Date`	Date of transaction (yy-mm-dd)
Account	`Sender_account`	Sender bank account ID
	`Receiver_account`	Receiver bank account ID
Transaction	`Amount`	Transaction amount (decimal)
	`Payment_type`	Payment type (e.g. Cash Deposit, Cheque)
Currency & Geography	`Payment_currency`	Sender account currency (e.g. UK Pounds, Euro)
	`Received_currency`	Receiver account currency (e.g. UK Pounds, Euro)
	`Sender_bank_location`	Sender bank location (e.g. UK, Spain)
	`Receiver_bank_location`	Receiver bank location (e.g. UK, Spain)
Label	`Is_laundering`	Binary indicator of laundering activity
	`Laundering_type`	Type of laundering pattern (e.g. Normal_Cash_Deposits, Normal_Fan_In, Smurfing, Over-Invoicing)

5.2.2 Class Imbalance

The dataset is highly imbalanced, with only 0.104% of transactions labelled as laundering. This reflects the nature of real-world AML data, where fraudulent activity represents a small fraction of all transactions. To take this class imbalance into account, AUPRC is used for model evaluation [3, 14].

5.2.3 Train/Test Split

The dataset will be split into a training set, validation set, and test set using a 70/15/15 ratio. This is inspired by the 70/30 split used by FraudGNN-RL [6], with the held-out 30% divided equally between validation and test to allow for hyperparameter tuning during training without leaking the final evaluation set. Given that the dataset contains a temporal dimension, all splits will be performed chronologically, ensuring the model is always trained on past transactions and evaluated on future ones. This approach avoids data leakage and better reflects real-world deployment conditions, where a model must generalize to transactions it has never seen before.

5.3 Data Risks

Several risks associated with the SAML-D dataset are acknowledged and should be considered when interpreting the results of this research.

Class imbalance. With only 0.104% of transactions labelled as laundering, the dataset is highly imbalanced. This may cause the model to be biased towards predicting the majority class, potentially missing actual laundering cases. This risk is mitigated by using AUPRC as the primary evaluation metric, which is specifically designed to handle imbalanced datasets.

Imbalance ratio. The class imbalance of 0.104% is notably low. ING reported that approximately 1–2% of transactions are flagged as potentially suspicious [3], though it is important to note that not all flagged transactions are confirmed laundering cases. Regardless, the dataset imbalance may still under-represent suspicious activity compared to real banking environments, which could affect model performance.

Synthetic data limitations. Although SAML-D is based on realistic AML patterns developed in consultation with AML specialists [16], it remains a synthetic dataset. Patterns present in real banking transactions may not be fully captured, and the model may not perform equally well when applied to real-world data.

5.4 Bias Consideration

Several sources of bias are relevant to this research and must be acknowledged.

Labeling bias. The labels in SAML-D are assigned based on predefined patterns developed in consultation with AML specialists [16], rather than verified real-world financial crime cases. This means the model may learn to detect the specific patterns encoded in the dataset rather than true laundering behaviour. Notably, this is not unique to synthetic datasets, as both ING and ABN AMRO acknowledged their labels are also derived from rule-based systems and human investigators, a process that is inherently error-prone [3, 14].

Historical bias. By training on past transaction patterns, the model may learn to replicate the flagging decisions encoded in the simulator rather than generalizing to new or unseen laundering strategies. As launderers continuously adapt their methods, a model trained on a fixed set of patterns may struggle to detect previously unseen patterns.

Class imbalance bias. With only 0.104% of transactions labelled as laundering, the model may be biased towards predicting the majority class. This risk is mitigated by using AUPRC as the primary evaluation metric and by applying threshold-based classification, allowing the decision boundary to be adjusted to improve recall at the cost of precision where necessary.

6 Model & Architecture

6.1 Proposed Techniques

This research proposes two models: a Deep Q-Network (DQN) as the Reinforcement Learning model and XGBoost as the supervised learning baseline. Both models will be trained and evaluated on the SAML-D dataset under the same conditions.

The DQN will be implemented following the architecture used by Qayoom et al. [17], which is the only pure RL approach found in the literature for transaction monitoring. The model will receive transaction features as its state, output a suspiciousness score as its action, and receive a reward based on whether the transaction is correctly classified. SHAP values will be applied post-hoc to interpret the model's decisions.

XGBoost will be implemented as the baseline model, trained on the same dataset using standard supervised learning. The model will output a continuous probability score per transaction, which will be used to set a detection threshold consistent with industry practice [3, 14].

6.2 Alternatives Considered

Several alternative models were considered before arriving at the final design choices.

For the baseline model, LightGBM and CatBoost were considered as alternatives to XGBoost. Both are state-of-the-art gradient boosting methods and were used by the top-performing teams in the IEEE-CIS Fraud Detection Competition [7, 4]. However, XGBoost was selected for three reasons. First, it is the most frequently used method in the AML literature, appearing in four of the studies surveyed by Bekach et al. [22]. Second, it is the primary model used in production at both ING and ABN AMRO [3, 14]. Third, it was used by all top-performing teams in the IEEE-CIS competition [7, 4, 18]. These three grounds make XGBoost the most defensible and practically relevant baseline choice.

6.3 Literature-based Justification

The choice of DQN is justified by the fact that both RL papers found in the literature apply a Deep Q-Network to transaction monitoring [6, 17]. Since no other RL architecture has been applied to this domain, DQN represents the most documented and directly comparable approach. The choice of XGBoost as the baseline is justified by the academic literature [22], industry practice [3, 14], and competition results [7, 4, 18], as discussed in the alternatives section. The use of SHAP is justified by its application in the AML literature [19] and its operational use at ABN AMRO [14].

6.4 Expected Trade-offs

Several trade-offs are anticipated between the proposed RL model and the XGBoost baseline.

Performance vs. adaptability. XGBoost is a well-established, highly optimised algorithm that is likely to achieve strong performance on the SAML-D dataset with minimal configuration. The DQN, by contrast, requires careful tuning of the reward function, learning rate, and exploration strategy, and may require more iterations to converge. However, the DQN has two theoretical advantages over XGBoost. First, its neural network architecture can capture complex, non-linear patterns in transaction data that tree-based models may not detect. Second, unlike XGBoost which is completely static after training, the DQN can in theory continue refining its policy through ongoing interaction with new transactions without requiring a full retrain.

Explainability. XGBoost is inherently more interpretable than a DQN, as feature importances can be directly extracted from the model. The DQN is a black-box neural network, and while SHAP values will be applied post training to provide explanations, it is not guaranteed that these explanations will be as clear or reliable as those produced by XGBoost.

Training complexity. XGBoost is straightforward to train and requires relatively little computational overhead. The DQN requires defining a suitable reward function, managing an experience replay buffer, and balancing exploration and exploitation, all of which add implementation complexity and computational cost.

7 Model & Architecture

7.1 Defined Metrics

This research uses the Area Under the Precision-Recall Curve (AUPRC) as the sole primary evaluation metric for both the DQN and XGBoost models.

7.2 Justification of Metrics

AUPRC is selected as the primary evaluation metric for three reasons. First, it is specifically designed for imbalanced datasets, where the minority class, in this case laundering transactions, represents only a small fraction of all transactions. Second, both ING and ABN AMRO use AUPRC as their primary evaluation metric in production [3, 14], making it the industry standard for AML transaction monitoring in the Netherlands. Third, it allows precision and recall to be evaluated simultaneously across all possible thresholds, which is critical in AML where the cost of missing a laundering case and the cost of generating too many false alerts must both be considered.

7.3 Validation Strategy

To ensure the reliability and reproducibility of the results, the following validation strategy is adopted.

The dataset is split chronologically into a 70% training set, a 15% validation set, and a 15% test set, inspired by the 70/30 split used by FraudGNN-RL [6]. The validation set is used for hyperparameter tuning during training, while the test set is reserved exclusively for final evaluation. The chronological split ensures that the model is always trained on past transactions and evaluated on future ones, preventing data leakage and better reflecting real-world deployment conditions.

To ensure reproducibility, all random processes, including model initialisation and any stochastic elements of the DQN training process, are controlled by fixing a random seed. Both models are trained and evaluated under identical conditions, using the same dataset, the same splits, and the same evaluation metric, ensuring a fair and reproducible experiment.

7.4 Stakeholder Evaluation Plan

The results of this research will be presented to De Lage Landen (DLL), a financial services company based in the Netherlands with an active AML department. As a financial institution subject to the Wwft [15], DLL has a direct interest in the effectiveness and explainability of AML detection models.

The evaluation will be presented in two parts. First, the AUPRC results for both the DQN and XGBoost models will be shared to give an objective view of model performance. Second, the SHAP-based explanations will be demonstrated to show how individual flagged transactions can be interpreted, addressing the explainability requirement imposed by the Wwft. Stakeholder feedback will be used to assess whether the model's outputs are interpretable and practically useful from an investigator's perspective.

8 Results

The results of this research will be reported once the experimental phase is completed. Success will be evaluated against the following criteria.

The primary success criterion is whether the DQN achieves a higher AUPRC score than the XGBoost baseline on the SAML-D test set under identical experimental conditions. As a secondary criterion, both models should aim for a recall of at least 0.95, consistent with industry practice at ING [3], though this is used as a reference point rather than a strict requirement. Finally, the SHAP-based explanations should be interpretable and meaningful, so they can be assessed during stakeholder evaluation with DLL.

It is important to note that if XGBoost outperforms the DQN, this is not considered a failure of the research. Rather, it would constitute a valid and informative finding, suggesting that supervised learning remains the more appropriate approach for AML detection on tabular data, which is itself a contribution to the literature given the absence of prior work comparing these two approaches in this domain.

9 Discussion

9.1 Anticipated Limitations

Several limitations are anticipated that may affect the findings of this research.

Synthetic dataset. The SAML-D dataset is synthetic, meaning the results may not fully translate to real banking environments. Patterns present in real transaction data may differ from those encoded in the dataset's.

Single dataset. This research evaluates both models on a single dataset. Results may therefore be specific to the characteristics of SAML-D and may not hold across different datasets or banking contexts.

DQN reward function design. The performance of a DQN is highly sensitive to the design of the reward function. A poorly designed reward may lead to suboptimal behaviour, such as the agent learning to avoid flagging transactions altogether to minimise penalties. Designing a reward function that accurately reflects the cost of false positives and false negatives in an AML context is a non-trivial challenge.

Explainability of DQN. While SHAP values will be applied post-hoc to interpret the DQN's decisions, it is not guaranteed that these explanations will be as reliable or consistent as those produced by XGBoost, which does not require additional explainability methods as its decision-making process is transparent by design.

9.2 Risk Analysis

The following risks are identified that could affect the successful completion of this research.

DQN convergence. Deep Q-Networks can be difficult to train and may fail to converge, particularly on highly imbalanced datasets. If the DQN fails to learn a meaningful policy, the comparison with XGBoost would not be informative. This risk is mitigated by closely following the architecture and hyperparameters used by Qayoom et al. [17] as a starting point.

Computational resources. Training a DQN on 9.5 million transactions may require significant computational resources. If training time becomes prohibitive, a smaller subset of the SAML-D dataset may be used, which could affect the generalizability of the results.

Class imbalance impact. The extreme class imbalance of 0.104% may make it difficult for both models to learn meaningful patterns. If neither model achieves satisfactory performance, the results may be inconclusive. This risk is mitigated by using AUPRC as the evaluation metric and applying threshold-based classification. If the imbalance proves too extreme during experimentation, the SAML-D data generator [16] can be used to generate additional laundering cases, reducing the imbalance to a more manageable ratio.

Stakeholder availability. The stakeholder evaluation with DLL depends on the availability of relevant personnel. If this evaluation cannot be conducted, several fallback options are available. As a first alternative, the evaluation will be conducted with a domain expert who has over 13 years of experience at ABN AMRO and holds an AML certification, providing equivalent industry expertise. If this is also not feasible, the evaluation will be conducted with a data scientist familiar with AML systems. In the worst case, the explainability analysis will be limited to a theoretical assessment based on the SHAP outputs alone.

9.3 Planned Validity and Reliability Safeguards

The following measures are planned to ensure the validity and reliability of the research findings.

Reproducibility. All experiments will be conducted with a fixed random seed to ensure that results can be reproduced.

Fair comparison. Both models will be trained and evaluated on the same dataset, using the same chronological 70/15/15 train/validation/test split and the same evaluation metric.

Transparent reporting. Both positive and negative results will be reported honestly. If XGBoost outperforms the DQN, this will be reported and interpreted rather than dismissed.

Literature alignment. All methodological choices are grounded in the existing literature and industry practice, as documented throughout this thesis. This ensures that the research design is defensible and comparable to prior work.

10 Conclusion

10.1 Clear Objective

This research investigates whether Reinforcement Learning can effectively detect money laundering and how it compares to leading supervised learning methods on tabular banking transaction data. To answer this, a Deep Q-Network is implemented and benchmarked against XGBoost, the current industry and academic standard, on the SAML-D dataset. The research additionally addresses the explainability requirement imposed by the Wwft by incorporating SHAP-based explanations into the proposed model.

10.2 Deliverables

This research produces the following deliverables:

- A trained XGBoost baseline model evaluated on the SAML-D dataset.
- A trained Deep Q-Network model evaluated on the same dataset under identical conditions.
- A quantitative performance comparison between both models using AUPRC.
- SHAP-based explanations for flagged transactions produced by both models.
- A stakeholder testing report based on feedback from DLL.
- A thesis document, serving as the full research report.

10.3 Realistic Planning and Milestones

The research is planned according to the following milestones:

Table 7: Planning and Milestones

Date	Milestone
23 March 2026	Ethics Peer Review presentation
1 April 2026	Progress meeting Block 3 and feedback processing
11 May 2026	70% version
4 May 2026	Progress meeting Block 4
13 May 2026	Feedback processing Block 4
12 June 2026	100% version

References

[1] Ismail Alarab, Simant Prakoonwit, and Mohamed Ikbal Nacer. Competence of graph convolutional networks for anti-money laundering in bitcoin blockchain. In *Proceedings of the 2020 5th International Conference on Machine Learning Technologies*, ICMLT '20, page 23–27, New York, NY, USA, 2020. Association for Computing Machinery. ISBN 9781450377645. doi: 10.1145/3409073.3409080. URL https://doi.org/10.1145/3409073.3409080.

[2] Erik Altman. Ibm transactions for anti money laundering (AML). https://www.kaggle.com/datasets/ealtman2019/ibm-transactions-for-anti-money-laundering-aml, 2019. Dataset available on Kaggle.

[3] Michael Attard and Louis de Bruijn. Machine learning binnen ING, March 2026. Guest presentation, ING Bank, Personal notes taken during presentation.

[4] Sergey Bryansky, Tony, krivoship, CPMP, and Giba. 2nd solution, CPMP view. https://www.kaggle.com/competitions/ieee-fraud-detection/writeups/2-uncles-and-3-puppies-2nd-solution-cpmp-view, 2019. Solution writeup for the IEEE-CIS Fraud Detection Competition on Kaggle.

[5] Dennis Cox. *What Is Money Laundering?*, chapter 1, pages 5–13. John Wiley Sons, Ltd, 2012. ISBN 9780470685280. doi: https://doi.org/10.1002/9780470685280.ch1. URL https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470685280.ch1.

[6] Yiwen Cui, Xu Han, Jiaying Chen, Xinguang Zhang, Jingyun Yang, and Xuguang Zhang. Fraudgnn-rl: A graph neural network with reinforcement learning for adaptive financial fraud detection. *IEEE Open Journal of the Computer Society*, 6:426–437, 2025. doi: 10.1109/OJCS.2025.3543450. URL https://doi.org/10.1109/OJCS.2025.3543450.

[7] Chris Deotte and Konstantin Yakovlev. 1st place solution - part 2. https://www.kaggle.com/competitions/ieee-fraud-detection/writeups/fraudsquad-1st-place-solution-part-2, 2019. Solution writeup for the IEEE-CIS Fraud Detection Competition on Kaggle.

[8] Europees Parlement en de Raad van de Europese Unie. Verordening (eu) 2016/679 — algemene verordening gegevensbescherming (AVG). https://www.privacy-regulation.eu/nl/artikel-9-verwerking-van-bijzondere-categorieen-van-persoonsgegevens-EU-AVG.htm, 2016. Van toepassing vanaf 25 mei 2018.

[9] FinCEN. What is money laundering?, n.d. URL https://www.fincen.gov/what-money-laundering.

[10] FIU-the Netherlands. What is money laundering?, 2023. URL https://www.fiu-nederland.nl/en/home/what-is-money-laundering/. Accessed: 2026-03-12.

[11] IEEE Computational Intelligence Society. IEEE-CIS fraud detection: Improve fraud prevention accuracy. Kaggle, 2019. URL https://www.kaggle.com/competitions/ieee-fraud-detection/overview. Accessed: 2026-03-12.

[12] Rasmus Ingemann Tuffveson Jensen, Joras Ferwerda, et al. A synthetic data set to benchmark anti-money laundering methods. *Scientific Data*, 10, 2023. doi: 10.6084/m9.figshare.c.6504421.v1. URL https://doi.org/10.1038/s41597-023-02569-2.

[13] Maryam Mahootiha. Money laundering data. https://www.kaggle.com/datasets/maryam1212/money-laundering-data. Dataset created as part of a Master's thesis. Generator available at https://github.com/mahootiha-maryam/moneylaundering-data-production.

[14] Smrithi Menon and Tin Nguyen. Aml at ABN AMRO, March 2026. Guest presentation, ABN AMRO Bank. Personal notes taken during presentation.

[15] Ministerie van Financiën. Wet ter voorkoming van witwassen en financieren van terrorisme (Wwft). https://wetten.overheid.nl/BWBR0024282/2026-01-01#Hoofdstuk3, 2008. Geraadpleegd op 14 maart 2026.

[16] Berkan Oztas, Deniz Cetinkaya, Festus Adedoyin, Marcin Budka, Huseyin Dogan, and Gokhan Aksu. Enhancing anti-money laundering: Development of a synthetic transaction monitoring dataset. In *2023 IEEE International Conference on e-Business Engineering (ICEBE)*, pages 47–54, 2023. doi: 10.1109/ICEBE59045.2023.00028. URL https://doi.org/10.1109/ICEBE59045.2023.00028.

[17] A. Qayoom, M.A. Khuhro, K. Kumar, M. Waqas, U. Saeed, S. ur Rehman, Y. Wu, and S. Wang. A novel approach for credit card fraud transaction detection using deep reinforcement learning scheme. *PeerJ Computer Science*, 10:e1998, 2024. URL https://doi.org/10.7717/peerj-cs.1998.

[18] Team Lions. 5th place solution: Lions. https://www.kaggle.com/competitions/ieee-fraud-detection/writeups/lions-5th-place-solution-lions, 2019. Solution writeup for the IEEE-CIS Fraud Detection Competition on Kaggle.

[19] Pavlo Tertychnyi, Mariia Godgildieva, Marlon Dumas, and Madis Ollikainen. Time-aware and interpretable predictive monitoring system for anti-money laundering. *Machine Learning with Applications*, 8:100306, 2022. ISSN 2666-8270. doi: https://doi.org/10.1016/j.mlwa.2022.100306. URL https://www.sciencedirect.com/science/article/pii/S2666827022000305.

[20] Mark Weber, Jie Chen, Toyotaro Suzumura, Aldo Pareja, Tengfei Ma, Hiroki Kanezashi, Tim Kaler, Charles E. Leiserson, and Tao B. Schardl. Scalable graph learning for anti-money laundering: A first look, 2018. URL https://arxiv.org/abs/1812.00076.

[21] Mark Weber, Giacomo Domeniconi, Jie Chen, Daniel Karl I. Weidele, Claudio Bellei, Tom Robinson, and Charles E. Leiserson. Anti-money laundering in bitcoin: Experimenting with graph convolutional networks for financial forensics, 2019. URL https://arxiv.org/abs/1908.02591.

[22] Bekach Youssef, Frikh Bouchra, and Ouhbi Brahim. State of the art literature on anti-money laundering using machine learning and deep learning techniques. In Aboul Ella Hassanien, Abdelkrim Haqiq, Ahmad Taher Azar, KC Santosh, M. A. Jabbar, Adam Słowik, and Parthasarathy Subashini, editors, *The 3rd International Conference on Artificial Intelligence and Computer Vision (AICV2023), March 5–7, 2023*, pages 77–90, Cham, 2023. Springer Nature Switzerland. ISBN 978-3-031-27762-7. URL https://rdcu.be/e8mZQ.